

Modes of Data Transfer and UART

1. Which mode of data transfer allows devices to transfer data without CPU intervention?

- A. Programmed I/O
- B. Interrupt-driven I/O
- C. DMA
- D. Polling

☒ **Answer:** C. DMA

2. In synchronous data transfer, data is transmitted:

- A. Without a clock signal
- B. With start and stop bits
- C. Along with a clock signal
- D. Randomly

☒ **Answer:** C. Along with a clock signal

3. Which technique checks device status repeatedly?

- A. Interrupt
- B. DMA
- C. Polling
- D. Buffering

☒ **Answer:** C. Polling

4. Interrupt-driven I/O improves efficiency because:

- A. CPU waits continuously
- B. CPU is interrupted only when needed
- C. It eliminates memory usage
- D. It uses no hardware

☒ **Answer:** B. CPU is interrupted only when needed

5. DMA transfers data between:

- A. CPU and ALU
- B. Memory and I/O device
- C. Registers only
- D. Cache and CPU

☒ **Answer:** B. Memory and I/O device

6. What does UART stand for?

- A. Universal Asynchronous Receiver Transmitter
- B. Unified Automatic Receiver Transmitter
- C. Universal Analog Receiver Transmitter
- D. Unique Asynchronous Transfer Tool

☒ **Answer:** A. Universal Asynchronous Receiver Transmitter

7. UART communication is:

- A. Synchronous
- B. Asynchronous
- C. Parallel
- D. Optical

☒ **Answer:** B. Asynchronous

8. Which bits are used in UART framing?

- A. Start bit
- B. Stop bit
- C. Parity bit
- D. All of the above

☒ **Answer:** D. All of the above

9. How many start bits are typically used in UART?

- A. 0
- B. 1
- C. 2
- D. 4

☒ **Answer:** B. 1

10. The baud rate in UART refers to:

- A. Voltage level
- B. Number of bits per second
- C. Number of bytes per second
- D. Clock cycles

☒ **Answer:** B. Number of bits per second